DARSI VAMSI
COMET FWC 071

Implementation of Three LED Blinking using VAMAN ARM Development Board

1 Objective

The objective of this experiment is to verify the General Purpose Input Output (GPIO) functionality of the VAMAN ARM Development Board by blinking three LEDs sequentially. The experiment demonstrates GPIO initialization, digital output control, and software-based timing delays.

2 Hardware Requirements

1. VAMAN ARM Development Board
2. USB Cable
3. Three On-board LEDs
4. Ubuntu 24.04 LTS System
5. GNU ARM Embedded Toolchain

3 Software Requirements

1. Ubuntu 24.04 LTS
2. GCC ARM Compiler
3. Make Utility
4. Serial Programming Tool
5. Makefile Build System

4 GPIO Configuration

The following GPIO pins are configured as digital outputs.

GPIO Pin	Function
GPIO 5	LED 1
GPIO 10	LED 2
GPIO 15	LED 3

Table 1: GPIO Configuration

5 Program Description

The application begins by configuring GPIO pins 5, 10, and 15 as output pins using the GPIO driver library.

Inside the infinite loop, each LED is turned ON one at a time while the remaining LEDs remain OFF. A software delay of 500 milliseconds is introduced after each LED transition, creating a continuous sequential blinking pattern.

The execution order is:

1. LED 1 ON
2. LED 2 ON
3. LED 3 ON
4. Repeat indefinitely

6 Algorithm

1. Initialize the GPIO subsystem.
2. Configure GPIO 5 as Output.
3. Configure GPIO 10 as Output.
4. Configure GPIO 15 as Output.

5. Turn ON LED connected to GPIO 5.
6. Wait for 500 ms.
7. Turn ON LED connected to GPIO 10.
8. Wait for 500 ms.
9. Turn ON LED connected to GPIO 15.
10. Wait for 500 ms.
11. Repeat the sequence continuously.

7 Program

```c
#include "gpio.h"
#include "delay.h"

void SystemInit(void)
{
}

int main(void)
{
    PyHal_GPIO_SetDir(5,1);
    PyHal_GPIO_SetDir(10,1);
    PyHal_GPIO_SetDir(15,1);

    while(1)
    {
        PyHal_GPIO_Set(5,1);
        PyHal_GPIO_Set(10,0);
        PyHal_GPIO_Set(15,0);
        delay_ms(500);

        PyHal_GPIO_Set(5,0);
        PyHal_GPIO_Set(10,1);
        PyHal_GPIO_Set(15,0);
        delay_ms(500);

        PyHal_GPIO_Set(5,0);
        PyHal_GPIO_Set(10,0);
        PyHal_GPIO_Set(15,1);
        delay_ms(500);
    }
}
```

```
32
33 void _start(void)
34 {
35     main();
36
37     while(1)
38     {
39     }
40 }
```

8 Testing Procedure

1. Connect the VAMAN ARM Development Board to the Ubuntu system.
2. Open the project directory.
3. Compile the project using the Makefile.
4. Generate the binary executable.
5. Flash the binary into the VAMAN ARM board.
6. Reset the board.
7. Observe the LEDs on the board.
8. Verify that the LEDs blink sequentially every 500 ms.

9 Expected Output

The LEDs should blink in the following sequence.

$$\text{LED 1} \rightarrow \text{LED 2} \rightarrow \text{LED 3} \rightarrow \text{Repeat}$$

Each LED remains ON for approximately 500 milliseconds before the next LED is activated.

10 Observation

- GPIO pins were successfully configured as outputs.
- LEDs responded correctly to GPIO HIGH and LOW signals.
- The software delay produced a visible blinking interval of approximately 500 ms.
- Only one LED remained ON at any instant.

11 Result

The Three LED Blinking program was successfully implemented and executed on the VAMAN ARM Development Board. The GPIO driver correctly controlled the three LEDs, and the sequential blinking verified the proper functioning of the GPIO interface, software delay routine, and program execution flow.

